

Mikhail Razakov

.NET Backend Engineer

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My professional experience is a C++/LLVM compiler internship at MCST. Backend evidence comes from independent ASP.NET Core projects with payment workflows, explicit state machines, idempotency, PostgreSQL/SQLite, reconciliation, and restart-recovery paths.

EXPERIENCE

July — August
2026

MCST — Compiler Engineering Intern

- Implemented an LLVM 22 LICM pass with loop analysis, side-effect and speculative-safety checks, and hoisting of loop-invariant instructions to a preheader.
- Developed a conservative interprocedural global-IV prototype with APInt affine evolution, transitive call effects, a separate legality phase, and an LLVM IR transform; unsupported CFG and call cases are left unchanged.

PROJECTS

CompilationLabLMS

An LMS with separate public/internal services, PostgreSQL as the source of truth, a ledger, payment orders, and a YooKassa flow that re-fetches provider state.

Working MVP: idempotent payment completion and persisted reconciliation converge on one completion path; deployment tooling is kept distinct from real production-readiness checks.

VpnMediator

A subscription/access service with a payment state machine, idempotency keys, an outbox/audit trail, and recovery workers.

The versioned deployment path coordinates backup of both databases, atomically switches releases, and restores the previous release after a failed health check.

UniversalToolchain

A large .NET project that separates planning from runtime and defines explicit contracts between independently developed modules.

The canonical test contract records 1,306 passing tests, with separate binding and interpreter/CIL parity checks.

SKILLS

.NET backend

C#/.NET, ASP.NET Core, REST/OpenAPI, webhooks, background services, provider integrations

Data & state

PostgreSQL, SQLite, transactions, state machines, idempotency keys, inbox/outbox

Failure handling

provider re-verification, reconciliation, retry/restart recovery, migrations, backup/restore

Delivery

Linux, Docker Compose, nginx, systemd, health checks, versioned releases, rollback

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

UniversalToolchain — Baltic Science and Engineering Competition

First-degree diploma and the Grand Prize “Perfection as Hope” for UniversalToolchain

MEPhI Junior

First-degree diploma, 96/100 for the UniversalToolchain project

Moscow / remote